

Raman Spectroscopy for Transparent Graphene Micro Electrode Arrays

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Introduction and Motivation

Raman spectroscopy is a widely used technique for characterizing carbon based nanomaterials such as carbon nanotubes and graphene. It is quick, non-destructive and accurate for determining the number of graphene layers. Here we present the assistance of this powerful technique in the fabrication of transparent graphene electrodes for micro electrode arrays (MEAs). MEAs are widely used to record the cellular activity from cells like neurons or cardiomyocytes. All typical standard electrode materials, Au, Pt, or TiN, are opaque, posing limitations for applications such as optogenetics and calcium imaging. Graphene offers the advantage of transparency, paving the way towards better simultaneous understanding of the cell-electrode interface and electrophysiology. Detailed correlative scanning electron microscopy (SEM) and micro Raman spectroscopy is performed confirming the presence of graphene after each processing step. SEM is used for visualization of graphene and Raman provides information about the number of layers and contamination, if any. Figure 1 shows a SEM image of a graphene conduction line and electrode. Figure 1 (b) and (c) are the 2D and G peak maps, respectively, over the area seen in (a). Graphene microelectrodes of 30 micron diameter are fabricated with the assistance of Raman spectroscopy and successfully tested with cell cultures.

MEA device

Micro Electrode Array (MEA)

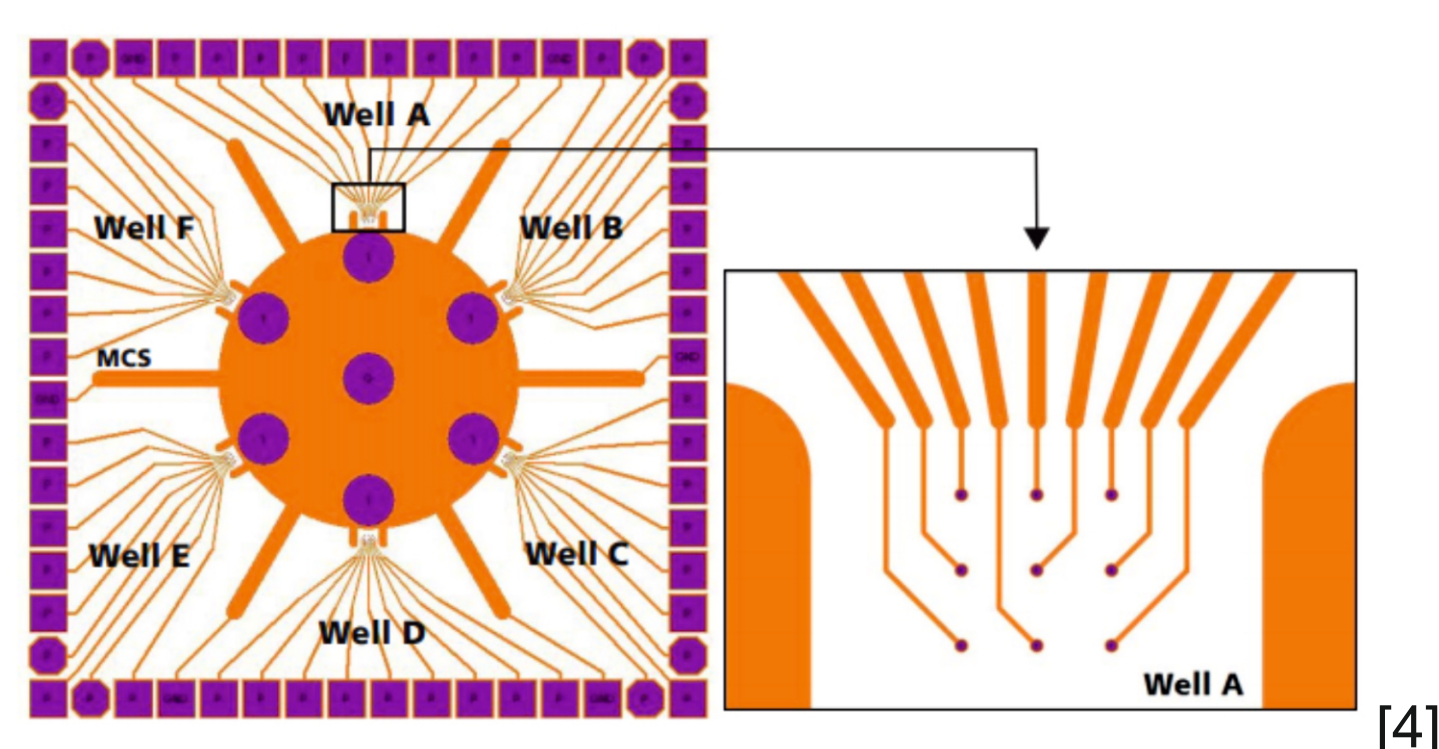


Figure 1: 6-well MEA layout with contact pads and conduction lines. Each well comprises 9 electrodes. The 6-well design offers the possibility to run 6 independent experiments simultaneously.

Graphene MEA Fabrication

Graphene processing scheme for integration in MEA

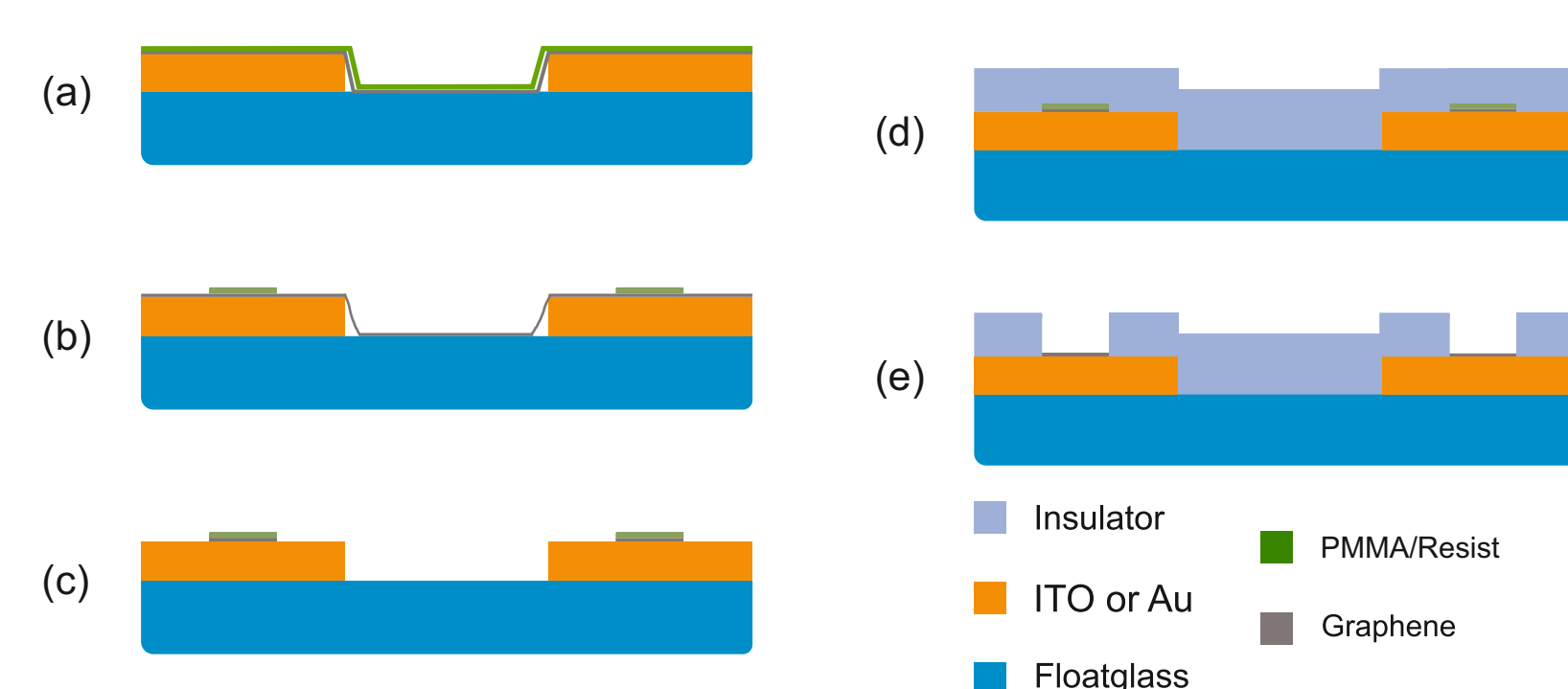


Figure 2: (a) Graphene transfer from Cu to Au conduction lines via PMMA stamp

- (b) PMMA removal followed by photolithography to structure the graphene by oxygen plasma
(c) Structured graphene with resist
(d) Insulator over the whole sample
(e) Opening insulator over the electrodes

Raman Characterization

The graphene Raman spectrum consists mainly of two peaks, namely, G (around 1580 cm^{-1}) and 2D (2680 cm^{-1}). In case of monolayer graphene, the peak intensity ratio 2D/G is two or higher. Furthermore, in the case of bilayer graphene, the 2D/G ratio is less than 2.

The Raman map was obtained with spectra taken with 300 nm spacing. Green laser was used with 1s acquisition time.

Figure 3: SEM image of graphene conduction track and electrode. 2D- and G-peak intensity maps affirm the presence of largely monolayer graphene. The defects in the graphene are also reflected in the maps.

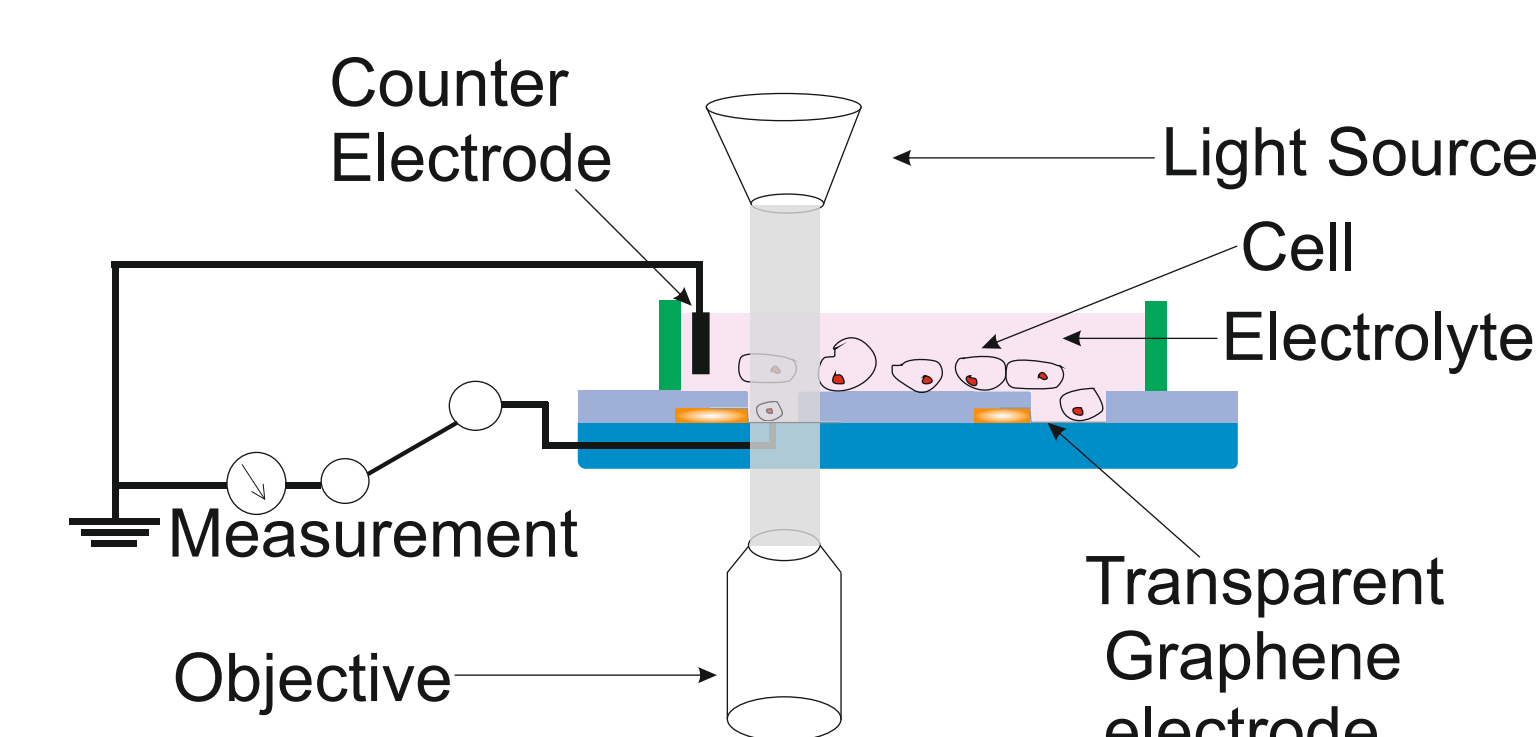
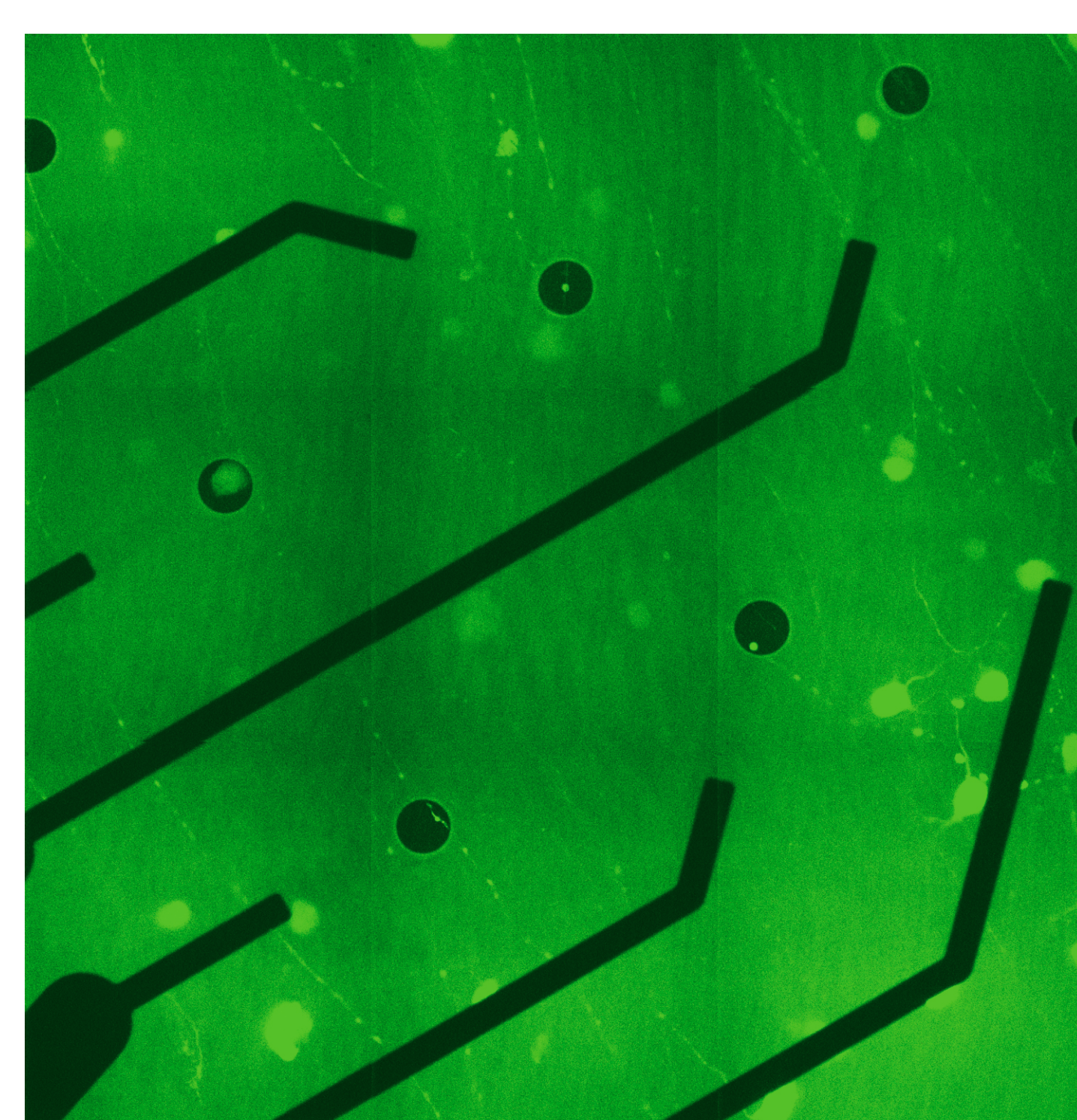
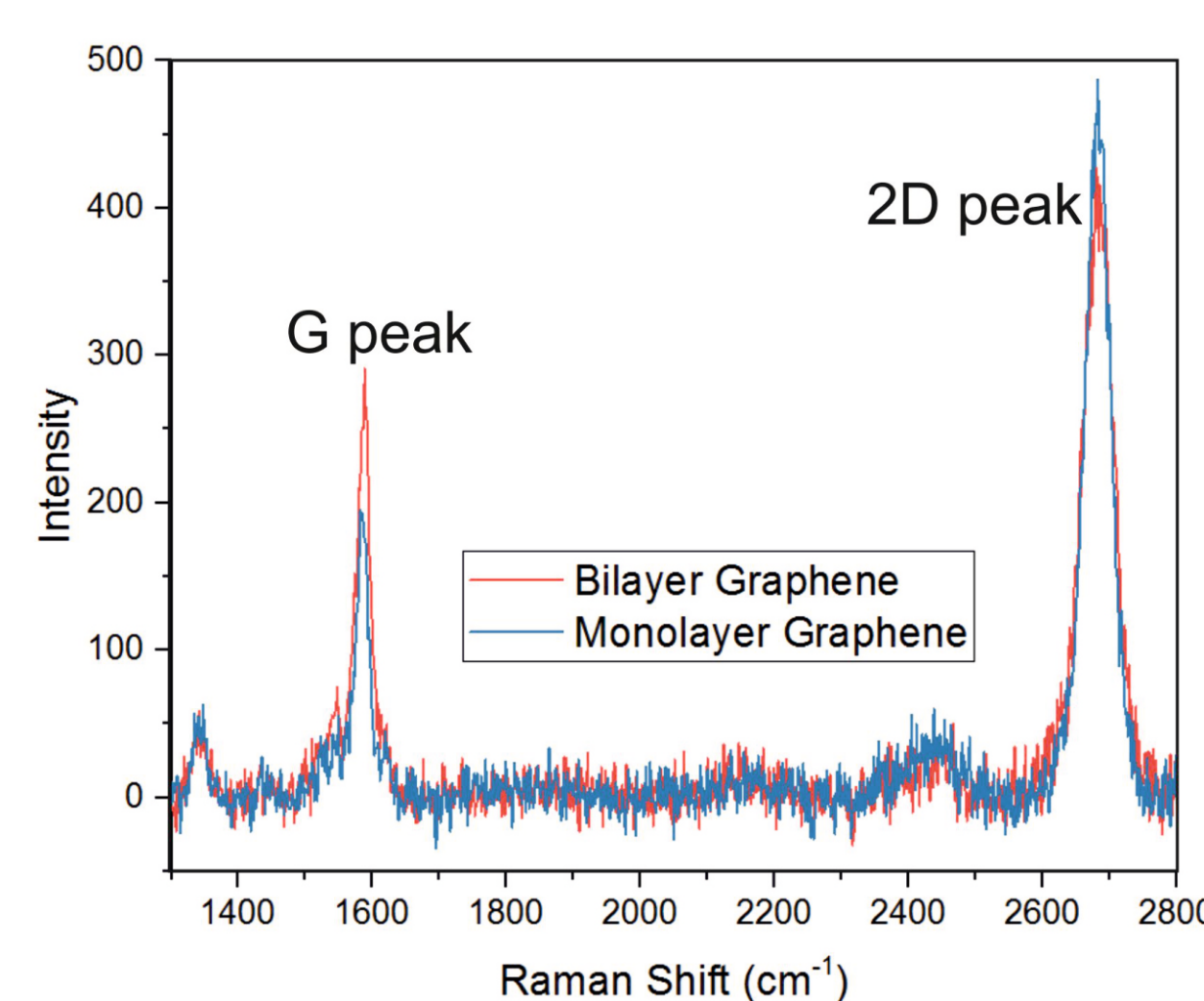


Figure 2: Schematic showing side view of MEA with cell culture on top and objective at the bottom. In the case of conventional electrodes, optical access to the electrode-cell interface is lost due to opaque electrode material. Hence, transparent electrodes need to be developed.

Results and Outlook

Reproducible and reliable graphene growth process is established following chemical vapour deposition with Cu as catalyst and methane as carbon source as seen in figure 4.

Largely monolayer graphene of size 2.5 cm x 2.5 cm is successfully produced and transferred to MEA conduction lines as seen in figure 5.

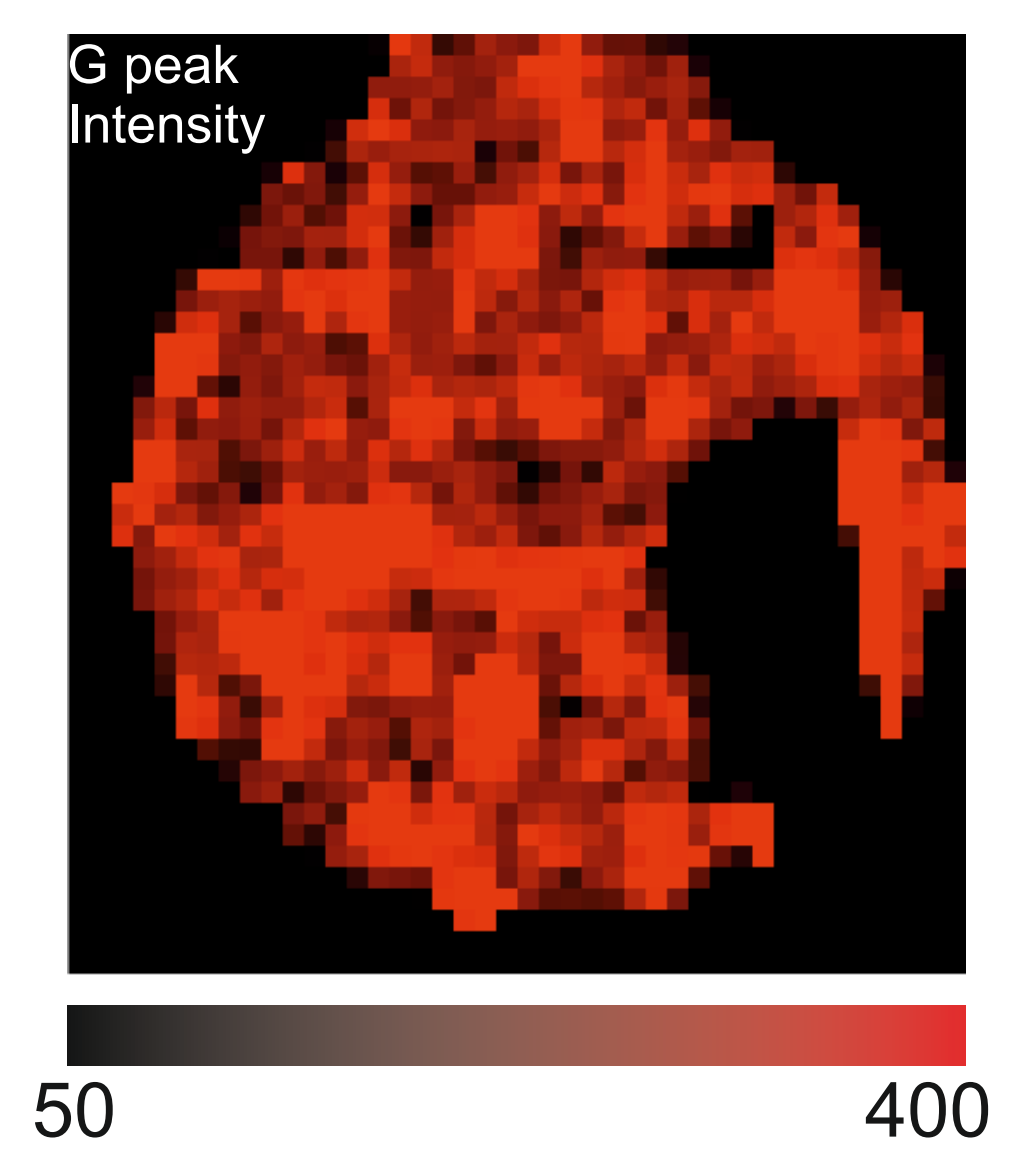
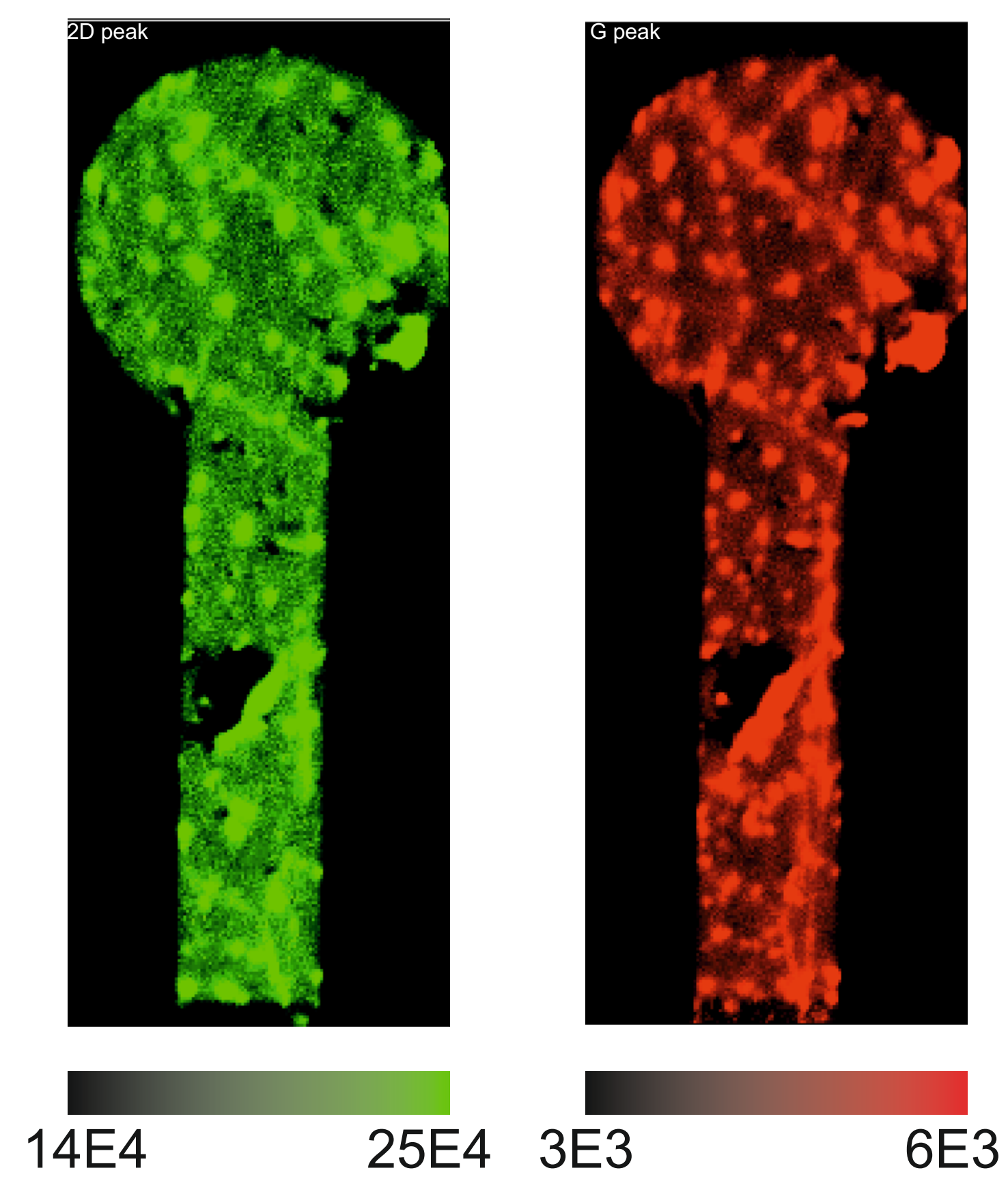
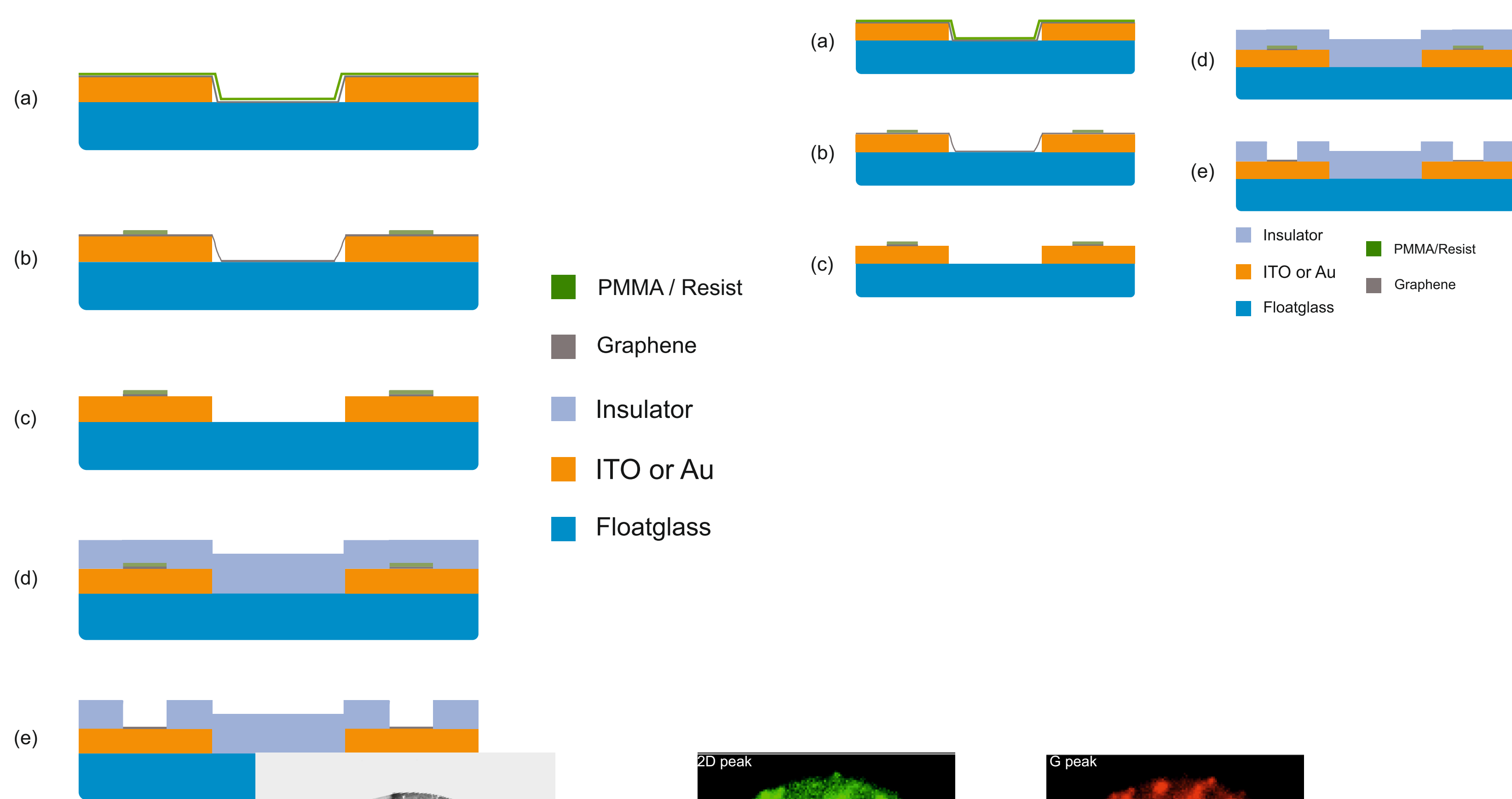
Graphene is structured by photolithography (figure 5) and the presence and quality of graphene is checked by SEM imaging and Raman mapping as seen in figure 7. The presence of largely monolayer graphene is confirmed by the Raman spectrum where G and 2D peaks are clearly visible.

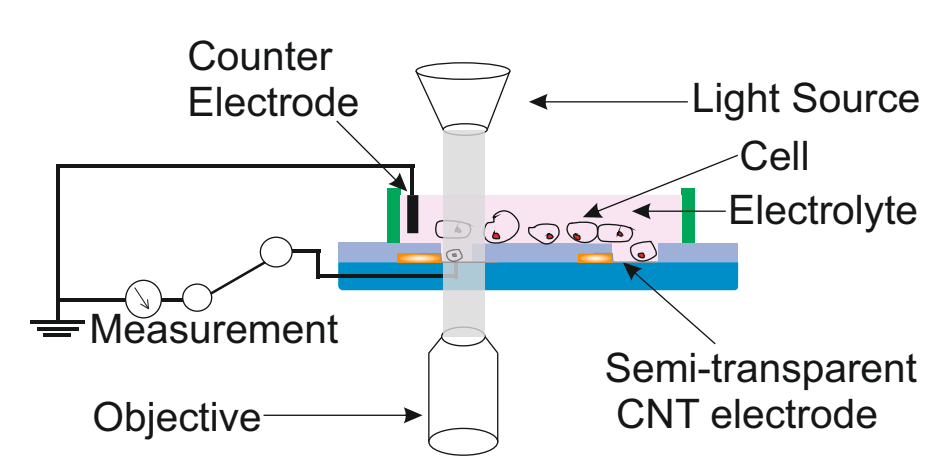
Experiments to test the applicability of the fabricated MEAs for optogenetic measurements and for calcium imaging are planned.

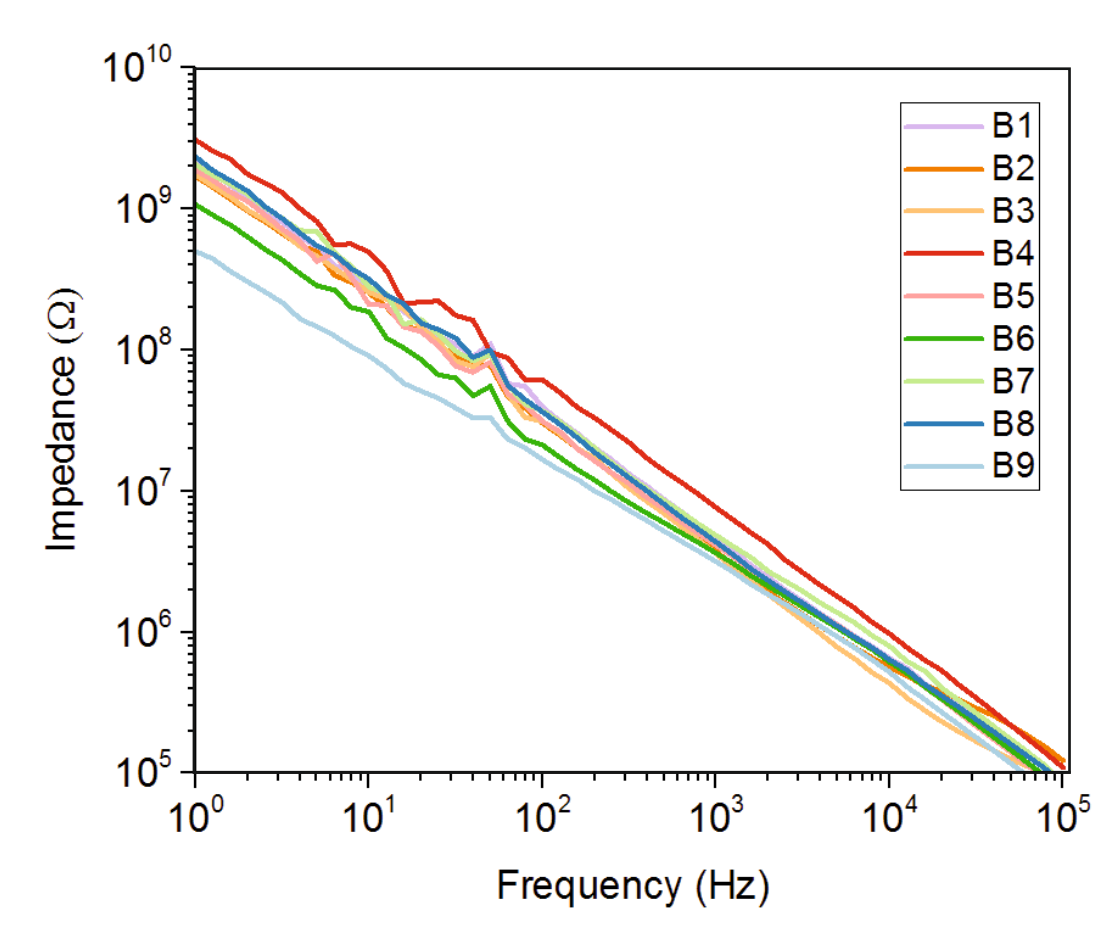
References / Acknowledgements

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- [3] D. Kuzum et al., Nat. Comm., 5, 5259 (2014)
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(a)



(b)



(c)



(d)



(e)



PMMA / Resist

Graphene

Insulator

ITO or Au

Floatglass